

Solution Requirements

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Input Handling	Allow a student to select a subject field and enter a free-text description of their learning problem (minimum 10 characters), with quick-example buttons for common cases.	High
2	Emotion Classification	Run the input through both a BiLSTM model and a fine-tuned BERT model and return the primary emotion with a confidence score for each.	High
3	Mixed Emotion Detection	Detect secondary emotions when their probability exceeds a threshold (0.15) and keyword-boost predictions using domain keyword lists.	Medium
4	AI-Generated Support	Send the field, problem, detected emotion and confidence to the Gemini 2.5 Flash API and return an empathetic, field-specific learning response.	High
5	Fallback Response	If the Gemini API call fails or is disabled, serve a predefined rule-based response (emoji, message, recommended action) for the detected emotion.	High
6	Interaction Logging	Persist every analysed interaction (field, problem, emotion, confidence, response, timestamp) to a CSV file for later review.	Medium
7	Analytics Dashboard	Visualise session history as an emotion distribution pie chart, an emotional-journey line chart, and a field-vs-emotion bar chart, plus summary metrics.	Medium
8	Session Controls	Let the user clear session history, toggle the use of Gemini, toggle CSV saving, and toggle display of analysis details.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Emotion prediction (BiLSTM + BERT) should return within a few seconds on CPU so the Streamlit UI stays responsive.	Combined inference under ~3-5 seconds on CPU
2	Scalability	The prediction pipeline should tolerate an increasing number of single-user sessions without code changes.	Stateless prediction functions per request
3	Security & Data Privacy	The Gemini API key must never be hard-coded; it is loaded from environment variables via python-dotenv.	API key stored only in a local .env file, excluded via .gitignore

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
4	Reliability & Availability	The app must keep working (via the rule-based fallback) even if the Gemini API is unreachable or errors out.	100% of requests return a response, AI or fallback
5	Usability & Accessibility	The interface should be understandable to a student with no ML background, using plain labels, emojis and progress bars.	Single-page flow, no technical jargon required to operate it
6	Maintainability	Prediction logic, preprocessing, logging and the Gemini helper are separated into individual modules under src/ for easy modification.	Each concern isolated to its own file under src/